

NAPLAN

2018 State report – Year 5

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Preface

State reports are issued by the QCAA about the performance of Queensland students on the National Assessment Program — Literacy and Numeracy (NAPLAN) paper tests. State reports provide system-level information and are publicly available. This report for Year 5 students in 2018 contains:

- the Queensland performance on each item
- the national performance on each item
- the item descriptors
- a commentary on the state results
- some recommendations for teaching.

Who should use this State report?

The NAPLAN State reports help principals, teachers and other school personnel understand, interpret and use information about student performance on NAPLAN.

School principals can use this report to provide information to the school community on aspects of the tests. This would allow professional conversations with their teachers, curriculum leaders, and department heads. Curriculum leaders can use this information to interpret the more specific information given in their school and class reports. These other reports are explained below.

Since this report is publicly available on the QCAA website, it can also inform providers of teacher training, special education services and educational research and policy.

Parents and carers can use this report to interpret the results on their child's student report. They are also able to judge how their child performed when compared with the whole population of students. The item descriptors provide them with useful information about the scope of the tests.

About the tests

The purpose of the National Assessment Program (NAP) is to collect information that governments, education authorities and schools can use to identify indicative literacy and numeracy skills Australian students can demonstrate. As part of that program, the NAPLAN tests are administered to full cohorts of students in Years 3, 5, 7 and 9 each year. These standardised tests are sources of information about student learning that can be used to inform educational policy and current educational practice.

The NAPLAN tests were initially developed using the nationally agreed Statements of Learning for English and Statements of Learning for Mathematics, 2005. Since 2016 however, the tests are referenced to the Australian Curriculum. The NAPLAN tests are designed to assess student understanding in the following areas:

- Language conventions: The test assesses the ability of students to independently recognise and use correct Standard Australian English grammar, punctuation and spelling.
- Writing: The test assesses the ability of students to convey thoughts, ideas and information through the independent construction of a written text in Standard Australian English.
- Reading: The test assesses the ability of students to independently make meaning from written Standard Australian English texts, including those with some visual elements.

- **Numeracy:** The test assesses students' knowledge of mathematics, their ability to apply that knowledge in context independently, and their ability to reason mathematically.

Marking and scoring the tests

Marking the tests

Markers mark those test items that do not use a multiple-choice format. These markers apply nationally agreed marking guides. There are marking guides for open-ended Reading items if any such items are included. Marking guides allow consistent and reliable judgments by markers. There are guides for the Writing test and one each for the constructed responses in Numeracy and Spelling. For some Numeracy items, students may provide a correct response in different forms. Professional officers decide on agreed scoring protocols for these items.

Calculating raw scores

The simplest calculation made in scoring the tests is the raw score — the number of questions answered correctly. All of the questions for the Language conventions, Reading and Numeracy tests are marked as either correct or incorrect. Raw scores for the Writing test are sums of the marks on each of ten criteria.

Raw scores have limited use. They enable the performance of students who have all completed the same test at the same time to be placed in a rank order, but they do not provide information about the level of difficulty of the test nor the relative differences between students.

Constructing scaled scores and bands

To make raw scores more useful, they are transferred to scores on a common scale that reflects how difficult it was to achieve each score. Each year ACARA publishes equivalence tables that allow a student's raw score to be located on the NAPLAN scale. The scale is comparable between year levels for each assessment area. An equating process is also carried out on each year's test to enable scores to be compared between successive years of testing. For example, a raw score of 20 on the Year 3 Reading test might be transformed to a scaled score of 354. This will also represent the same achievement for a student with the same scaled score in Year 5, and for a student with the same scaled score for Reading in a previous year.

Each NAPLAN scale is divided into ten bands used to report student progress.

Using NAPLAN reports to inform teaching and learning

Using scaled scores

The scaled score can compare the results of different students. Scaled scores provide a basis for measuring and comparing students' abilities across years of schooling, for example comparing a student's result in Year 3 in 2016 and Year 5 in 2018. The scales can thus help to monitor the growth of groups of students over time. This enables the school to review and/or consolidate special programs that may have been put in place.

Principals and teachers should take care when making comparisons between small groups of students. For groups of fewer than 10 students, differences may not be reliable, particularly small differences.

Using item analysis

While the national and state reports provide the comparative data, class reports provide a school with the information that can be used to inform teaching and learning and to build capacity in schools. Analysis of the NAPLAN class data, in particular the performance on each item, will

provide teachers with information about the understandings and patterns of misunderstandings of students.

Looking at the performance on the items and then analysing the error patterns allows teachers and principals to make hypotheses about why groups of students make particular errors. As mentioned below, more detailed analysis by QCAA staff is available from the QCAA website.

Steps for analysis might be as follows:

- Compare the facility rates (percentage correct) achieved by the school's students with the national and state results available in this document. Is their performance consistent?
- Look at the common errors made by their students and compare them with the common errors made in the state. Only errors from Queensland students are available and are found in the item analyses that are part of SunLANDA Online.
- Form hypotheses about why students are making these errors, e.g.
 - How did students think about this aspect of the curriculum?
 - What misunderstandings might these errors represent?
 - How might the structure of the test question have shaped the response?

Using a combination of the NAPLAN data, school data and professional judgment, teachers may then test these hypotheses to see whether they are valid or whether there is more to be thought about and investigated. Teachers can then plan lessons related to the general areas where students seem to need help. Teachers can also make judgments about teaching approaches and curriculum.

The professional conversations that are part of this process are the most effective and powerful way to use the data, as they are the vehicle for developing shared understandings.

Placing the tests in the assessment context

The results from the NAPLAN tests should be seen as only one input into a school's assessment program. Various forms of assessment are needed to inform the different stages of the teaching and learning cycle. Principals and teachers should keep in mind that NAPLAN is a point-in-time, timed test that can only cover a few curriculum features.

The results from a school's own assessments of students should be consistent with the NAPLAN test results. If the test results are different from what was expected, consider possible reasons. The results of the tests may indicate aspects of student performance that need further investigation within the classroom, using other forms of assessment.

An item with a low facility rate (percentage correct) may not necessarily indicate a problem in teaching and learning. It may be that this was simply a difficult item for all students in this cohort across Australia.

Other NAPLAN reports

In addition to the State reports, the following reports are produced about the performance of Queensland students who sit the NAPLAN paper tests:

SunLANDA Online

Since 2015, student data has been released on the QCAA School Portal using the SunLANDA Online interface. Access to SunLANDA as application software is also still available on the QCAA website.

SunLANDA Online provides class and school information in an electronic form that permits customised spreadsheet generation by users. In addition, it shows representative samples of students' incorrect responses to constructed responses where applicable. Hyperlinks from within SunLANDA Online lead to the QCAA's test item analysis. Information on how to use this service is available at: www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda.

Test item analysis

These PDF documents contain an analysis of each test item. They can be downloaded directly from the QCAA website: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school Brief Identification Code (BIC) and password is required to access these documents. The analysis reproduces each item followed by expert analyses of how the item operated. It shows the distractors presented in multiple-choice items and explains students' reasoning.

School and class reports

The NAPLAN school and class reports are supplied electronically on the secure section of the QCAA website. These reports are accessible only with the school's Brief Identification Code (BIC) login and password. Individual student reports are distributed to schools as printed copies.

School reports

The QCAA issues NAPLAN school reports giving information about each school's performance. They provide a summary of year-level performance as well as performance by gender, language background and Indigenous status in the following fields:

- distribution of scaled scores
- distribution of achievement bands
- school and state means
- participation of the group.

The school report positions a school's performance within the state on a graph that is shaded to show the range of performance for the middle 60% of Queensland students together with the state mean.

Class reports

The QCAA issues NAPLAN class reports that show the performance of every student on every item. Under the name of each student is recorded the items they had correct and incorrect. They also show students' responses to constructed-response items.

The class report also gives the:

- percentage correct for each item for the class and state, and by gender

- scaled scores for each student
- performance bands for each student.

Individual student reports

The QCAA issues individual student reports to schools after the tests. Schools receive one printed report for each student to distribute to parents/carers.

ACARA reports

As well as the Queensland reports from the QCAA, national reports are available from the website of the Australian Curriculum Assessment and Reporting Authority (ACARA). The *NAPLAN National Summary Report* and the *NAPLAN National Report* allow states and territories to place the achievement of their students in relation to their peers across the nation. This is system-level information and is publicly available.

Literacy

Writing

Stimulus (writing prompt) Years 3 & 5

YEAR 3 AND YEAR 5

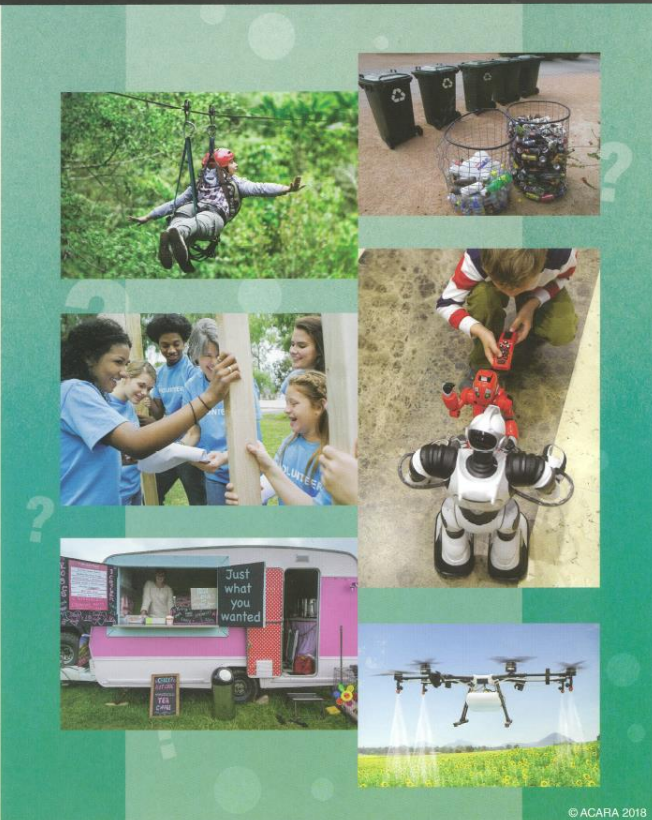
My big idea!

Write a persuasive text about an idea that would improve your classroom, school or community. Give reasons why it is a good idea by explaining how it might be useful and what its benefits are. You can use your own idea or you can use an idea on this page.

- **Start with an introduction.**
An introduction lets a reader know what you are going to write about.
- **Write your opinions on the topic.**
Give reasons for your opinions. Explain your reasons.
- **Finish with a conclusion.**
A conclusion sums up your reasons so that a reader is convinced of your opinions.

Remember to:

- plan your writing
- use paragraphs to organise your ideas
- write in sentences
- choose your words carefully to convince a reader of your opinions
- pay attention to your spelling and punctuation
- check and edit your writing.



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About the task

In 2018, the NAPLAN Writing test was based on the persuasive genre. Two prompts were used, one for Years 3 & 5 and another for Years 7 & 9. The test conditions and administration remained the same as in previous years, i.e. teachers delivered the same spoken instructions and read the text aloud to students. Working independently, students had to plan, compose and edit a written response. Students were allowed five minutes to plan their script, thirty minutes to write and a further five minutes to edit and complete the task. Three pages were provided for students to write a response.

The 2018 prompt for Years 3 & 5 was titled *My big idea!* Students were asked, in the textual component of the prompt, to *Write a persuasive text about an idea that would improve your classroom, school or community.* The prompt was relatively open-ended, allowing students to *use your own idea or you can use an idea on this page.* Six photographic images were provided for the students and the majority of Year 5 students selected one of these. A large number of students used the idea of rubbish and recycling rubbish to make their schools or community cleaner, or a zip-line to make school more fun. Interestingly, a number of students who selected their own idea chose the idea of having a class pet to teach students to be responsible and to reduce stress in the classroom.

Additional information identified the structural components of a persuasive text and further defined these elements, e.g. *Start with an introduction. An introduction lets a reader know what you are going to write about.* Other notes were also provided in relation to the conventions associated with a writing task, e.g. *write in sentences, use paragraphs to organise your ideas, pay attention to your spelling and punctuation, check and edit your writing.* If students are made aware of the information given on the prompt it will give them clear instructions about what they need to do to produce a complete persuasive text.

Markers for this Writing test were trained using the national persuasive writing marker training package, delivered as part of ACARA's national assessment program. Markers were recruited and trained in accordance with national protocols. For Queensland, registered teachers marked the NAPLAN Writing test. All markers applied the 10 criteria and related standards from the marking rubric. Writing test scripts were marked on screen in all states and territories. Stringent quality-control measures were applied to the marking of student scripts, including a prescribed percentage of scripts to be double-marked, and the daily application nationally of control scripts for all markers. As part of the Queensland marking operation for 2018, referee marking further ensured marker reliability. There was also provision for appeal over individual Writing test scores after the results were released.

A copy of the Persuasive Writing Marking Guide (2013) is available at:
www.nap.edu.au/_resources/Amended_2013_Persuasive_Writing_Marking_Guide_-_With_cover.pdf.

Performance

Anecdotal evidence from markers indicated that students in Years 3 and 5 were comfortable with the writing prompt, *My big idea!* The images provided were widely adopted by students as the basis for their persuasive texts, particularly the photo of the different rubbish bins and the zip-line. A 'tour the stimulus' approach was less evident in 2018 than in some previous years. The relatively small group of students who diverged from the images provided on the prompt tended to write more challenging texts, though this was not exclusively so. A closer adherence to the suggestions provided on the stimulus page appeared to be reflective of a general appropriateness for the age levels, rather than some absence of originality of ideas.

In 2018 students continued to adopt formulaic approaches to this persuasive text type, and whilst this broadly met the demands of the task (and rubric), it restricted the natural flow of language and ideas. So, typically, an opening statement, such as *I firmly* (or *strongly* or even *passionately*) *believe*, would be followed by a series of body paragraphs, often introduced by the connectives *Firstly, Secondly, Thirdly* — concluding with a paragraph that provided a brief summation of arguments and ideas presented.

Students in Year 5 generally wrote in straightforward sentence forms, with complex sentences common, but reflecting a limited range, e.g. *because, if*, etc. As far as possible, students should be encouraged to explore the full range of sentence structures, including the judicious use of short, sharp simple sentences and fragments for effect.

In terms of length of text, students in Years 3 and 5 tended to write a little more than the persuasive texts of previous years. This was encouraging. It is possible that the suitability of the prompt contributed to this trend. Obviously, students who were able to make use of a well-chosen adverb, adjective or nominal group could write with fewer words but with more demonstrable control of language.

A number of students used more varied openings to their texts to engage the reader. In some cases, this was productive. Following are three examples of introductions written by Year 5 students in Queensland.

Every day the world is changing. We've gone from box TVs to hover boards in less than a century. Schools need to set up their students for success, but how can they if they can't embrace technology like ipads? If their students don't have the knowledge of devices, then they are going to fail in the real, changing world. (Year 5 Student 1)

I am one-hundred percent confident, that by reducing the amount of waste we bring to school, our community will be a much cleaner place! As a young guardian of the environment, I believe it's an excellent idea to have rubbish free days at school. Just a Monday and Friday, without any rubbish, could make a huge difference! Waste free days, are just what all schools need. (Year 5 Student 2)

Imagine you were taking your family to the aquarium (sic). You show them the wonders of the tanks and the decorations in them. But you can't show them the marine life because they have all died from choking on plastic bags that they thought were jellyfish or other food. Is that what you want the future to look like in just a couple of years. (Year 5 Student 3)

In other cases, the introductions appeared a little contrived, and failed to match the development of ideas (and tone) in the remainder of the text. Whilst cohesion is only one of the NAPLAN Writing criteria, the ways in which ideas, vocabulary, paragraphing and sentence formation work together is important to whole-of-text integration and reader understanding. Students should be encouraged to focus on 'big' ideas and construct their texts accordingly, ensuring consistency and sincerity of message without unnecessary gimmickry.

Relative areas of improvement from 2017 included the criteria of *Persuasive devices*, *Text structure*, *Paragraphing*, and *Spelling*. When completing the 2018 Writing test, the majority of Year 5 students successfully demonstrated an understanding of the structure of a persuasive argument and a range of persuasive devices to use in order to persuade a reader to a point of view on their chosen *Big idea*.

Smart Robots Should be Used in Everyday Life

Imagine the scenario. The life of a patient is in your hands. There is one way to save him (or her) – with a drug delivered directly into the liver. But how can you do that? With nano sized robots! of course! This is a situation starting to appear, and many also believe that robots should also be used in our everyday communities too. Will you start to believe too?

First of all, scientists and deep sea explorers all over the world are experimenting with special robots to recover ship wrecks. In addition, they're also being used to map previously unknown parts of the seafloor. This prevents the risk of humans getting crushed by the immense water pressure of the deep. Do you want to support these incredible creations?

In addition, A.I robots can also be used to save lives. Back to our situation in the opening paragraph, the surgeon has just brought in the tray containing the tiny robots. He then injects them into the blood stream. As the robots head for the liver at a speed of 1mm per second, which is supersonic for a robot invisible to the naked eye. They inject the drug and go dormant, and the bots are flushed out in urine. In total, the operation takes half an hour and the patient suffers no side effects! So now what do you think of these heroes of the future?

Finally, on the news these days, people often lose lots of jobs, which the government either eliminates or the person has bad education. Robots could solve both problems. By teaching kids about Artificial Intelligence, they learn how to use it. When they're older, they could get jobs building robots or using them in assorted ways. Do you want to help your child's growth?

So, overall, it has been clearly stated that robots should become a solid part of our lives. There are many reasons like education, exploration, life saving and much, much more. So, bring on the robots and artificial intelligence today!

END OF TEST

Commentary on sample script

The majority of students who approached the Years 3/5 prompt tended to use the images provided on the prompt page as their subject matter. The student who wrote the sample script chose the picture of the robot and then wrote about different types and uses for robots and the advantages for all.

In the introduction, the writer set the scene with an example referred to in more detail in the third paragraph. He concludes the paragraph with a direct address to the reader:

... many also believe that robots should also be used in our everyday communities too. Will you start to believe too?

This direct address to the reader is a persuasive device used to be inclusive of the reader, drawing them to identify with the writer's position.

It is quite difficult for younger students to escape the fairly formulaic approach to a persuasive text. Limitations in vocabulary, breadth of subject knowledge and familiarity with the genre restrict the capacity for Years 3 and 5 students to explore more original text forms. While this persuasive text does follow a fairly traditional approach to a NAPLAN demand writing task, the writer has managed to 'colour' the text with an original voice, as well as demonstrate a control over the development of their thought process about the need for further development and use of robots.

So, overall, it has been clearly stated that robots should become a solid part of our lives. There are many reasons like education, exploration, life saving and much, much more. So, bring on the robots and artificial intelligence today!

The student's position in support of robots is clearly established in the introductory paragraph, which states the topic and the position taken. Following on from the introduction, the student uses three body paragraphs to persuade readers to their point of view, sharing examples of the uses of robots and the many benefits. In terms of the interpersonal elements, the writer actively includes the reader throughout the argument, e.g. *Will you start to believe too?; Do you want to support these incredible creations?; So now what do you think of these heroes of the future?; Do you want to help your child's growth?* At the end of each paragraph the writer directly addresses the reader so that the reader is continually challenged to think about the examples given.

Throughout the text the student extols the virtues of different types of robots. The young writer carefully structures his text to initially engage readers, direct them to his stance, thoughtfully sift through supporting arguments involving the many ways in which robots could make our lives better, then firmly restates his position for a sense of closure.

One of the real strengths of this text is the command of appropriate language demonstrated by the writer, together with a sense of tone and writer's voice. Mature phrases — e.g. *the patient suffers no side effects; experimenting with special robots to recover ship wrecks; used to map previously unknown parts of the sea floor; prevents the risk of humans getting crushed by the immense water pressure (sic) of the deep; By teaching kids about Artificial Intelligence, they learn how to use it* — illustrate not only a command of vocabulary, but a capacity to elaborate and extend the meaning of that vocabulary selection.

The text scored highly in Paragraphing, with consistent use of topic sentences and paragraphs that logically sequence the development of the argument. Sentences demonstrate variety in form, length, and complexity, and flow naturally because of thematic choices, connectives and conjunctions.

Punctuation, with just a minor slip or two, helps pace and direct the reader through the text and spelling is almost flawless. Overall, the text provides a quality exemplar of this text type.

Language conventions

Spelling

Results and item descriptions

The percentage columns give the facility rate (percentage correct).

Item	Answer	QLD %	Aust. %	Description
Proofreading — Error identified				
1	sister	95.16	95.79	Corrects the spelling of a two-syllable word ending in -er.
2	shrug	83.99	85.43	Corrects the spelling of a single-syllable word ending in a vowel and consonant.
3	snail	74.67	77.55	Corrects the spelling of a single-syllable word with ai.
4	microphone	57.64	62.91	Corrects the spelling of a three-syllable word starting with micro-.
5	imaginary	50.33	53.73	Corrects the spelling of a multisyllable word ending in -ary.
6	misjudged	40.76	40.74	Corrects the spelling of a two-syllable word with dge.
7	approval	37.01	39.76	Corrects the spelling of a three-syllable word with prov.
8	mixture	37.32	39.99	Corrects the spelling of a two-syllable word with x.
9	machinery	30.58	31.58	Corrects the spelling of a multisyllable word with the suffix -ery.
10	excitement	37.56	39.68	Corrects the spelling of a three-syllable word with the suffix -ment.
11	awkwardly	31.30	31.43	Corrects the spelling of a three-syllable word starting with awk-.
12	article	23.96	24.51	Corrects the spelling of a three-syllable word ending in -le.
Proofreading — Error not identified				
13	recount	77.83	80.36	Identifies and corrects an error in a two-syllable word with ou.
14	shirt	78.91	79.86	Identifies and corrects an error in a single-syllable word with ir.
15	crooked	68.42	69.46	Identifies and corrects an error in a two-syllable word starting with cr-.
16	cancel	58.33	60.54	Identifies and corrects an error in a two-syllable word with ce.

17	delivered	61.81	62.89	Identifies and corrects an error in a three-syllable word starting with del-.
18	weapons	46.11	49.26	Identifies and corrects an error in a two-syllable word with ea.
19	radiates	26.37	29.34	Identifies and corrects an error in a three-syllable word ending in -ates.
20	spectacular	25.96	28.96	Identifies and corrects an error in a multisyllable word ending in -ar.
21	withered	23.51	26.48	Identifies and corrects an error in a two-syllable word ending in -ed.
22	courtesy	20.64	22.67	Identifies and corrects an error in a three-syllable word ending in -esy.
23	contents	18.91	22.17	Identifies and corrects an error in a two-syllable word ending in -ents.
24	sympathetic	10.20	12.67	Identifies and corrects an error in a multisyllable word starting with sym-.
25	rejoiced	6.78	7.89	Identifies and corrects an error in a two-syllable word with oi.

About the test

The 2018 Year 5 test focused on the following spelling features:

- short vowels and the schwa sound: *sister*, *shrug*, *shirt*, *article*, *weapons*
- long vowels and diphthongs: *snail*, *awkwardly*, *recount*
- consonants: *crooked*, *misjudged*, *mixture*, *cancel*
- inflections: *radiates*, *withered*
- affixes: *microphone*, *imaginary*, *machinery*, *excitement*, *spectacular*, *courtesy*, *sympathetic*
- base words and roots: *approval* (from prove), *rejoiced* (from joy), *contents* (from *tenere*), *delivered* (from *liber*).

Both sets of questions in the Spelling test use proofreading formats. The target words in the first set are misspelt and identified by being circled. Those in the second set are misspelt but not identified, so students have to find them inside sentences containing other words (distractors) that are correctly spelt. These supplied misspellings may lead students to spell differently when they write to dictation or compose their own sentences.

Performance

Compared to the national average, Year 5 students in Queensland performed only marginally lower on all words except *misjudged*. Compared to previous Queensland cohorts, Year 5 students performed similarly in Spelling to recent years.

About 10% of students omitted any response to the last three words. The vocabulary demand in these low-facility words is very high.

Very many students did not attempt to spell the target words *rejoiced* and *spectacular* because they selected distractor words instead. Similarly, the provided misspellings of the words *approval* and *article* may have influenced many of the student errors.

Often, students' errors show that they rely inappropriately on 'sounding out' spelling strategies, especially when confronted with less familiar words, e.g. *withered* and *rejoiced*. In the case of the word *misjudged*, however, where sound is a guide to spelling, many students simply accepted the supplied misspelling of the word *misjudged* as 'misjudged'.

Implications for teaching

Test-wiseness

Ignore supplied errors: Students should avoid being influenced by the supplied misspelling of a word. Instead, they should focus on their own spelling knowledge and strategies.

Meaning

Students need to learn how chunks of letters can convey grammatical meanings. These chunks include plural inflections and tense inflections (e.g. *radiates*, *withered*) and affixes (e.g. *microphone*, *imaginary*, *machinery*, *excitement* and *spectacular*). Exposure to a rich vocabulary, in both spoken and written forms, is an aid to various literacy skills. When students hear the correct pronunciation and contextual use of an advanced word, they gain knowledge of its meaning and likely spelling.

Beyond 'sounding-out'

Phonemic awareness: Students have to learn how the sounds match letters alphabetically in many single-syllable words. Students need phonemic awareness or the ability to hear all the sounds inside a word and they need alphabetic knowledge of the sounds of each letter. Next, patterns containing more than one letter are learned, such as the digraph *oi* for the diphthong sound and the C*Ve* pattern whereby a long vowel sound is shown by a final *e*.

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school BIC and password is required to access each year level document.

Grammar and punctuation

Results and item descriptions

The percentage columns give the facility rate (percentage correct).

Item	Answer	QLD %	Aust. %	Description
26	A	96.51	96.75	Identifies the subordinating conjunction that correctly completes a complex sentence.
27	A	87.79	89.28	Identifies the correct use of the article an with a word starting with h-.
28	D	88.40	89.52	Identifies a sentence that requires a question mark.
29	A	85.89	88.30	Identifies a word that cohesively links two parts of a complex sentence (synonymy).
30	D	76.35	77.27	Identifies the sentence boundary between two simple sentences.
31	C	73.00	74.01	Identifies the correct use of capital letters for a geographical place name.
32	B	65.23	67.63	Identifies that a sentence has narrative text features.
33	B	65.62	66.67	Identifies the correct use of clause commas in a complex sentence.
34	A	71.02	71.73	Identifies the correct verb to ensure subject-verb agreement in four sentences.
35	A	63.33	64.11	Identifies that a sentence expresses an opinion.
36	D	77.36	78.41	Identifies the sentence which correctly combines information from two sentences.
37	B	62.35	63.00	Identifies a contraction that grammatically matches a sentence.
38	B	53.67	55.65	Identifies the correct use of a connective to create a coherent sentence.
39	C	45.89	47.49	Identifies the sentence boundary between two sentences.
40	D	41.37	40.98	Identifies the action verb in a simple sentence.
41	C	35.75	36.46	Identifies the reason for the use of an apostrophe in a simple sentence.
42	D	30.77	30.10	Identifies the order of events in a complex sentence.
43	C	29.57	28.94	Identifies correct location of quotation marks for direct speech.
44	B	28.65	27.53	Identifies the subordinate clause in a sentence.
45	B	26.16	25.55	Identifies a sentence that lacks a main clause.

46	C	14.35	13.53	Identifies an adverbial phrase in a sentence.
47	B	21.92	19.74	Identifies the sentence using contractions correctly.
48	D	16.64	14.61	Identifies the abstract noun in a simple sentence.
49	D	21.44	21.68	Identifies the incorrect noun/pronoun agreement in a complex sentence.
50*	2,1,1,1	21.92	21.72	Identifies the correct subject/verb agreement in four sentences.

* Item 50. Answers in sequence: were, was, was, was.

About the test

The NAPLAN Language conventions items test sentence-level, clause-level and word-level skills. The test does not cover the curriculum. Instead, it tells how a large number of students perform on a small range of tasks. Standardised tests can, however, suggest broad trends across a cohort. At the level of individual students, NAPLAN results can supplement classroom assessments and guide teachers to important points of grammar and punctuation that need revisiting. The Language conventions test comprises 50 items in total: 25 Spelling items and 25 Grammar and punctuation items.

The Grammar and punctuation items in the 2018 Year 5 test included the following features:

- Cohesion: Students had to follow the causal and sequential links within sentences created by connectives (Item 38) and synonymy (Item 29).
- Sentence structure: Students had to identify appropriate punctuation to demarcate sentences (Items 28, 30 and 39). Students were also required to identify subordinating conjunctions (Items 26, 36 and 44), sequence of events (Item 42) and subject-verb agreement (Items 34 and 50).
- Apostrophes: Students had to know how to use contractions correctly in context (Items 37 and 47). They also had to recognise a possessive plural apostrophe (Items 41 and 47).
- Commas: Students had to identify correct clause commas (Item 33).
- Direct speech: Students had to know the punctuation of and the typical sentence structures for interrupted speech attribution (Item 43).
- Adverbs: Students had to identify the adverbial phrase within a sentence (Item 46).

For information about the full range of grammar knowledge that Year 5 students should have, refer to the Australian Curriculum English.

Performance

The performance of Queensland Year 5 students in Grammar and punctuation was comparable with the national cohort. Queensland student results were closer to the National average in this strand in 2018, compared with the other strands of Spelling, Reading and Numeracy that were also assessed in a multiple-choice format. Between 2008 and 2012, Queensland student results were below the National average, however this gap closed steadily over this period. In both 2013 and 2014, the difference between Queensland and National was slight. Queensland performance has mirrored the National average from 2015 to the current year's results.

The strongest results by Queensland students were achieved in Item 44 (subordinate clauses), Item 46 (adverbial phrases), Item 47 (contractions) and Item 48 (abstract nouns). It is interesting

to note that Queensland Year 3 students also performed above the national average in items dealing with contractions.

The weakest results by Queensland students were in Item 27 (articles), Item 29 (synonymy), Item 32 (textual features), Item 38 (connectives) and Items 28, 36 and 39 (sentence boundary punctuation).

A notable feature of the performance of Queensland Year 5 students on this year's test was the relative strength of results in the final set of items, which tended to be the most difficult. Relative to national averages, Queensland students performed higher in eight of the final ten items on the test.

Implications for teaching

Grammar and punctuation is not a separate area of reading and writing, but an essential component. The contexts of language in use are pivotal to a full understanding of how syntax, grammatical and punctuation conventions operate. Wherever possible, meaning and purpose in texts should be emphasised, noting how particular language conventions contribute to making meaning.

NAPLAN tests grammar and punctuation at the level of single sentences, but this should not be the only way to teach or assess these skills in the classroom. Rather, teaching how a sentence fits into a wider text will influence choices about the sentence's pronouns, its verb tense, its order of components (subject, verb and object) and its elaborations.

Familiarity with diverse types of texts may help students to be more confident in viewing the NAPLAN items. Teachers are encouraged to guide students through notable grammar and punctuation features and usage in a wide selection of reading materials, including texts that are challenging and divergent in form.

Metalinguage

Success in the Grammar and punctuation items of the 2018 Year 5 Language conventions test relied to some extent on a degree of metalinguistic knowledge and understanding. This year there were nine items requiring metalinguistic knowledge, a similar number to 2017 (nine items) and 2016 (ten items).

Terminology such as *action verb* (Item 40), *adverb*, *verb*, *adjective*, *noun* (Item 48), *adverbial phrase* (Item 46), *apostrophe* (Item 41), *subordinate clause* (Item 44) and *main clause*, *subject* and *connective* (Item 45) can reasonably be expected to continue to feature within NAPLAN Language conventions tests.

The teaching of these terms is essential for many reasons other than NAPLAN success. A common language between teacher and student, student and student and even test constructor and student is an enabler for student language improvement. However, knowledge of grammatical terms, by itself, is not sufficient for students' language growth. Students should be encouraged to see language in context, to look at linguistic concepts within the frame of making meaning, and even to recognise how parts of speech can fulfil different roles. At Year 5 level, the foundational concepts introduced in the earlier years of schooling should be consolidated and reinforced regularly in the written texts students encounter and create.

Proofreading

Students need practice in proofreading so that they are able to proofread their own work and recognise errors in grammar, weaknesses in style or inappropriate grammar for a specific genre. It is also important to expose students to exemplary texts and point out how a sentence can be crafted, balanced, given pace and rhythm and contribute to the tone or meaning of the whole text.

Apostrophes

Many students struggle to understand the concepts of contraction, possession and plurality and how they relate to spelling and apostrophes. It is a valuable skill for a student to clearly know and apply the rules governing contractions and possession in the early years.

Commas

The commas that separate a subordinate clause from the main clause obey simple conventions. Students should learn about this as they learn how and why the subordinate clause can be placed in different positions in relation to the main clause.

Direct speech

The tense sequence and the use of first and third person pronouns for reported and direct speech follows conventions that students need to learn (e.g. She said '**I will** attend' / She said that **she would** attend).

Test-wiseness

It is pleasing to note that the omission rates for Grammar and punctuation are noticeably lower than those for Spelling, Reading and Numeracy. Even for the more difficult items towards the end of the test, the omission rate did not exceed 4%. Students should continue to be encouraged to provide responses for all items in multiple-choice format questions when they are checking over their responses. Persistence to provide responses to all items and a systematic approach to answering and checking over answers will contribute to the omission rate remaining low in Grammar and punctuation.

Item 35 presented students with five multiple-choice options, rather than the more conventional four options. Item 50 was presented in a similar format to previous years and required students to choose between two options in each of four given sentences. The increased complexity and demands in these items may influence their lower facility rates. It is important that students understand that items may be presented in unfamiliar ways and that reading questions carefully is an important component of test-wiseness.

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school BIC and password is required to access each year level document.

Reading

Results and item descriptions

The percentage columns give the proportion of correct answers (facility rates).

Item	Answer	QLD %	Aust. %	Description
<i>Letter from camp</i>				
1	C	96.85	97.07	Locates directly stated information in a personal letter.
2	D	76.34	76.59	Links directly stated information across paragraphs in a personal letter.
3	C	97.10	97.10	Locates directly stated information in a personal letter.
4	A	84.14	85.52	Interprets directly stated information in a personal letter.
5	C	81.57	81.27	Identifies the purpose of descriptive language in a personal letter.
6	A	67.69	68.36	Identifies the main purpose of a personal letter.
<i>The procession</i>				
7	C	85.75	86.08	Identifies the setting for a narrative.
8	A	49.27	50.50	Evaluates a character trait from behaviour in a narrative.
9	CD	49.60	49.51	Locates directly stated information in a narrative.
10	B	57.83	58.05	Locates directly stated information in a narrative.
11	B	55.85	55.90	Identifies the purpose of a paragraph in a narrative.
12	D	42.32	41.73	Interprets directly stated information in a narrative.
<i>Why we should care about Antarctic krill</i>				
13	A	60.13	61.93	Identifies the main idea of a persuasive discussion.
14	A	70.31	70.59	Interprets a pronoun reference in a persuasive discussion.
15	D	52.03	52.00	Identifies contrasted information in a persuasive discussion.
16	B	35.58	35.36	Interprets directly stated information in a persuasive discussion.
17	B	75.62	78.54	Identifies the author's perspective at the end of a persuasive discussion.
18	C	58.17	58.33	Identifies the main idea of a persuasive

				discussion.
19	A	74.98	77.57	Identifies the author's intent in a persuasive discussion.
<i>Wave-driven generator</i>				
20	B	67.49	68.54	Identifies the purpose of a diagram in an information text.
21	A	67.79	68.23	Interprets a diagram in an information text.
22	A	72.56	73.12	Interprets directly stated information in an information text.
23	A	38.48	38.05	Interprets directly stated information in an information text.
24	D	33.85	35.56	Identifies the purpose of a subheading in an information text.
25	D	37.41	37.78	Analyses information to make an inference in an information text.
26	D	41.82	43.21	Identifies a difference between an information text and a persuasive text.
<i>Titanic</i>				
27	B	42.23	42.45	Identifies the purpose of the text layout in an advertisement.
28	D	17.67	18.88	Interprets a device used to engage readers of an advertisement.
29	D	33.06	34.38	Identifies evaluative language to match a theme in an advertisement.
30	B	34.24	36.57	Interprets vocabulary that contributes to the tone of an advertisement.
31	B	25.65	27.13	Identifies directly stated information in an advertisement.
32	C	21.86	22.07	Identifies an author's assumption in an advertisement.
33	A	23.96	23.69	Identifies the author's perspective in an advertisement.
<i>Australia's rarest insect goes global</i>				
34	D	15.98	16.43	Links information from the text to the title of an information text.
35	AD	4.09	3.90	Identifies how persuasion is used in the opening paragraph of an information text.
36	A	37.39	37.90	Infers the reason for an outcome in an information text.
37	C	47.95	48.80	Locates directly stated details in an information text.

38	C	49.54	50.71	Interprets the meaning of figurative language in an information text.
39	C	26.91	27.05	Identifies the significance of an event in an information text.
40	B	31.52	31.74	Interprets directly stated information in an information text.

About the test

In 2018, the Year 5 Reading test consisted of 40 items based on six reading magazine units:

1. Letter from camp
2. The procession
3. Why we should care about Antarctic krill
4. Wave-driven generator
5. Titanic
6. Australia's rarest insect.

The Year 5 Reading test has the greatest number of link units. Four of the reading units are links to either Year 3 or Year 7. *The procession* and *Why we should care about Antarctic krill* link to the Year 3 Reading test. *Wave-driven generator* and *Titanic* link to the Year 7 Reading test.

Most items had a standard four-option, multiple-choice format. However, Item 9 (The procession) requires two keys selected from five options. Item 35 requires the student select **all** that apply. Because the number of keys is unspecified, the item difficulty is greatly increased. Typically, these item formats increase difficulty. Item 35 is the most difficult in the Year 5 Reading test.

Items generally increase in difficulty as students progress through the test. Facility rates decline as students progress from the first reading unit to the last. Reading demands generally increase, distractors become more sophisticated and item construction becomes more complex.

Rather than categorise the reading units by genre, it is more useful to categorise the items by question type (see table below). This is partly because many of the reading stimulus texts contain features of multiple genres. Furthermore, reading the individual item analysis, the error analysis sometimes shows that students will select a distractor because it alludes to a specific genre. Even capable students will be drawn to these distractors.

Categorising items by *question type* puts the focus on the more relevant reading strategies. For example, an examination of the question types makes it apparent that simply underlining or highlighting text may only be useful for one of the seven question types — Literal-recall questions. See the *Implications for teaching* section in this text, and SunLANDA for a range of reading strategies that are specific to the types of questions.

	Question type:	The reader is asked to:
Literal	Recall	Recognise or recall information.
	Translation	Change information into a different form — it might involve paraphrasing the ideas or restating them in terms or forms other than those in the text.
Text-based inferential	Interpretation	Identify the relationships among ideas, definitions, facts and values — these would involve such relationships as comparisons and cause and effect; they involve a minimum of higher-order thinking as the reader/learner needs only to respond to and manipulate ideas in the text.
Higher-order (Context-based) inferential	Application	Solve real-life problems by extrapolating what is in the text — readers/learners need to combine ideas from the text with prior knowledge.
	Logical analysis	Analyse and judge the quality of the logic inherent in the text — readers/learners might, for example, identify fallacies or points of view represented in a text.
Creative	Synthesis	Respond to a problem or idea with original and creative thinking.
	Evaluation	Make judgments with respect to specific criteria.

Performance

The average facility rate for the first unit, *Letter from camp*, is 83.95%. The average facility rates for the last two units declined to 28.38% and 30.48% respectively. *Letter from camp* is the most accessible text and most of the question types are literal. There was one question, Item 5, that required context-based inference as it asked the reader to identify the purpose of descriptive language.

The average facility rate was much reduced to 56.82% in the second unit, *The procession*. The most difficult item in this unit was Item 8, which required the reader to identify and evaluate a character trait from the character's behaviour in the narrative. Making this more challenging was the fact that two of the options were highly plausible. Another reason this was the most difficult item in the Reading test so far was that higher order inferential questions typically require reading beyond the sentence level to the whole text level.

In the third reading unit, *Why we should care about Antarctic krill*, the most difficult item had a 35.58% facility rate. This was unusual because it was a literal-translation question (which usually has a higher facility rate). The complexity of this item was due to the density of the text and the use of tiers 2 and 3 words. In addition, difficulty was increased because the reader had to interpret the options. To illustrate this, many capable students selected a distractor because they misinterpreted the meaning of this distractor rather than the meaning in the text.

In *Wave-driven generator* (link to Year 7), the average facility rate is 12.96%, lower than the average for Year 7 students in this unit. Again, the most difficult item was a higher-order inference question, Item 24. In this item, the reader had to identify the purpose of a subheading. Typically students focus at the sentence level, however inferential questions, and particularly higher-order inference questions, require reading the entire text.

Titanic is the second of the link units with the Year 7 Reading test. This unit had the lowest average facility rate in Year 5 of 28.38%. Three of the items required higher-order inference skills. The average Year 5 facility rate for this unit was 16.43%, lower than the average Year 7 facility rate. The greatest difference was in Item 30, which required higher-order inference. Year 5 students performed comparatively well in the final sequencing item in this unit (Item 33).

The most challenging item in this unit for Year 5 students was Item 28 which required that they *interpret a device used to engage readers*. It is likely that what students found challenging was the stem referred to the *opening sentence*, however, as is the case with any higher-order or context-based inference, the student needs to read this in the context of the entire text.

In the final reading unit, *Australia's rarest insect goes global*, all of the facility rates were 50% or less. The most difficult item was Item 35 with only a 4% facility rate. As well as requiring higher-order inference, this was not presented in a standard multiple-choice format; students are instructed to *Choose **all** that apply*.

Furthermore, five options are provided, all of which require careful reading and solid vocabulary knowledge. Of the two correct options, one was more obvious as the reader would expect it to be about *triggering the reader's curiosity*. The other option was more difficult to access as the reader had to understand that the author uses language in a deliberate way to engage the reader.

In summary, Year 5 students found inferential questions, particularly higher order inference more challenging. Students require reading strategies that help them to access the overall message of the text.

Implications for teaching

This year, Year 5 students demonstrated the ability to correctly answer many literal (recall and translation) type questions. However, reading strategies that limit students to reliance on keywords and literal recall will not equip them with the skills required to understand the overall message of a text. As a result, items requiring students to make higher-order inferences continued to be challenging, suggesting that students need to develop strategies to help them make inferences as they read, i.e. to draw conclusions about the unknown based on the known.

As a general note, all questions involving purpose, main idea, theme or tone of the text (in whole or part) challenge students because they have to understand the whole of the text to answer the question. The big challenge for teachers is to get students to read a variety of texts, annotate them in the classroom and discuss the ideas in the texts in groups so that they can see how all the parts of the text contribute towards the meaning of the whole. This is the time to discuss patterns in the text (e.g. cause and effect, contrast, comparison), identify connections between ideas in the text as well as the two or three main parts of the text and how the parts contribute to the overall meaning. All of this should occur before students begin a close study of the text. Students will handle the distractors in the questions much better if they are clear about the subject matter and the purpose of the text **before** they proceed to the questions. The QCAA website provides teacher advice on specific strategies for annotating and discussing challenging texts with a focus on helping students make inferences.

Teachers need to encourage students to read for pleasure and recreation to extend their knowledge of themselves and the world around them. Reading develops empathy for characters and people in difficult situations. Students also need to be able to confidently participate in a close study of a text, to check for fallacies and persuasive techniques, and to draw attention to emotive language and literary techniques. World citizens need to be discerning and capable readers and confident speakers and writers about those texts.

The complexity of the reading process becomes apparent when students discuss texts and share how they arrive at their personal understanding of the text. Students need to experience using extracts in which not everything in the text is clear e.g. sometimes a character's motivation is not clear, and students should not panic if the subject matter is not completely accessible.

Teachers are the facilitators of this process of annotating and discussing texts; they are not the leaders. Teachers should focus on:

- modelling a love of books and reading
- finding authentic texts at an appropriate level which will engage their students
- promoting higher-order questioning of texts (both set texts for special study and unseen texts for close study)
- reading aloud to students to promote reading for pleasure
- talking about texts and authors respectfully and disagreeing with each other about their interpretations appropriately (Where is your evidence for that?)
- developing an awareness of how the parts of the text combine to create a whole through both semantic (links between the ideas) and syntactic (grammatical links) cohesion
- encouraging students to make inferences as they read (i.e. an informed guess supported by evidence from the text)
- encouraging students to see connections between the text and their own knowledge and experience, between different things within the text and between this text and other texts in a similar genre or on similar subject matter. This will assume increasing importance as students approach secondary schooling
- encouraging students to be active readers and make connections between the text and their own knowledge, experience and feelings
- providing students with opportunities to discuss and share their understandings of a text while always asking them to substantiate their interpretations from the text.

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school BIC and password is required to access each year level document. Additionally, further advice and support can be found in: QCAA 2015, *Beyond NAPLAN: How to read challenging texts*, Beyond NAPLAN series: www.qcaa.qld.edu.au/downloads/p_10/naplan_read_challenging_texts.pdf.

Numeracy

Results and item descriptions

The Numeracy strands from the Australian Mathematics Curriculum are abbreviated as follows: Number and Algebra (NA); Measurement and Geometry (MG); Statistics and Probability (SP).

All items are worth one score point. However, there is a range of difficulty across the items.

The percentage columns give facility rates (percentage correct) for each item.

Item	Strand	Answer	QLD %	Aust. %	Description
1	NA	B	95.18	95.09	Recognises the value of Australian coins.
2	MG	A	92.68	93.45	Compares the capacity of familiar objects.
3	NA	75	77.12	78.00	Makes the largest two-digit number given three possible digits.
4	SP	38	83.44	83.58	Counts the total from a tally table.
5	NA	14,6	82.98	83.06	Uses the connection between addition and subtraction to write a number sentence.
6	SP	A	87.02	86.57	Chooses a suitable scale to describe the data on a column graph.
7	NA	2	61.91	64.12	Determines a missing number to make a number sentence true.
8	NA	C	79.45	80.50	Continues a number pattern involving repeated addition or multiplication.
9	MG	3,2	71.74	74.24	Identifies the faces used to make a prism.
10	MG	B	84.70	84.23	Uses directional language to describe a route on a basic map.
11	NA	A	61.28	63.77	Calculates the change required for a simple transaction.
12	SP	B	69.65	71.96	Identifies an event where one cannot happen if the other happens.
13	MG	B	74.74	74.18	Identifies a shape after a rotation.
14	SP	A	75.53	73.78	Selects appropriate methods for data collection.
15	MG	C	76.80	79.22	Compares and estimates objects using metric units of length.
16	NA	7	71.09	75.11	Solves a simple division problem.
17	MG	C	61.13	64.05	Counts the number of blocks to calculate the volume of cubes.
18	NA	C	56.52	58.83	Solves a multistep problem involving addition and subtraction using a number line.

19	SP	AC	56.23	56.88	Interprets a simple two-way data table.
20	NA	D	49.60	52.04	Solves a problem involving division resulting in a remainder.
21	NA	B	66.07	69.78	Identifies a quarter of a whole subdivided into 8 parts.
22	NA	360	47.29	51.38	Applies place value knowledge to solve a subtraction problem.
23	SP	7	49.55	50.92	Interprets picture graphs where one picture represents many data values.
24	NA	C	41.07	44.33	Solves a two-step word problem involving multiplication and division.
25	NA	A	40.52	41.98	Calculates the length of an object by subtracting decimal metric units.
26	MG	C	45.19	48.83	Converts litres to millilitres to solve a capacity problem.
27	MG	8	42.40	44.86	Uses scaled thermometers to compare temperatures.
28	SP	AE	48.19	50.71	Identifies possible outcome of rolling a dice once.
29	MG	D	34.55	36.34	Uses a timetable to solve a problem involving duration of time.
30	NA	17	29.13	30.61	Estimates total cost by rounding amounts to the nearest dollar.
31	MG	AD	35.98	39.83	Compares the area of irregular polygons.
32	NA	AD	23.47	23.87	Select two multiplication number sentences to represent a combinations problem.
33	NA	BC	27.17	29.40	Identifies expressions which could be used to continue a number pattern.
34	NA	D	29.39	32.59	Represents a problem as a number sentence involving multiplication and addition.
35	SP	C	21.81	23.89	Interprets a side-by-side column graph to compare categorical variables.
36	MG	8	19.94	20.98	Compares angles within complex shapes and identifies those that are acute.
37	NA	D	18.05	20.35	Represents numbers to three decimal places on a number line.
38	NA	BCE	9.82	13.05	Identifies factors of a number and uses them to solve a problem.
39	NA	AE	10.28	12.02	Identifies the five-digit numbers that satisfy a given rule.
40	MG	A	10.90	12.11	Uses a calendar to determine the day of the week of a particular date.

41	NA	BF	12.20	13.34	Identifies equivalent fractions used in context.
42	NA	31.5	7.00	9.27	Calculates the cost of an item after a percentage discount.

About the test

The Year 5 NAPLAN Numeracy test covered concepts and skills from across the Australian Mathematics Curriculum. The NAPLAN Numeracy test consists of 42 items covering concepts and skills from three content strands. Student results are reported as a single score.

The distribution of the 42 items across the Australian Mathematics Curriculum strands was:

- Number and Algebra (22 items)
- Measurement and Geometry (12 items)
- Statistics and Probability (8 items).

Over the entire test of 42 items, 30 (71%) were multiple-choice, with the remaining 12 items (29%) requiring students to arrive at their own answers (constructed responses). The number of constructed responses was slightly higher than in previous years. Interpretation of tables, graphs or diagrams was required in 26 of the 42 items (62%). The remaining 16 items (38%) were word problems that usually incorporated numerals within the information and question stems.

Performance

Overall, Queensland students performed comparably with the national cohort. This year, 96.1% of Queensland Year 5 students scored at or above the national minimum standard (Band 4), compared to 95.6% of Australian students overall.

The performance by Queensland students on five test items (Items 1, 6, 10, 13 and 14) were above the Australian facility rates (percentage correct). In all other items, the difference between the Queensland and the national facility rates was less than 4%.

Reflecting the gradual increase of difficulty used in standardised tests, facility rates lowered as the test proceeded. For example, 95% of Queensland students correctly answered Item 1, 66% correctly answered Item 21 (the half-way mark of the test) and only 7% correctly answered Item 42. On constructed response items, Queensland students were less successful than the national cohort in all instances.

Items of interest, where the Queensland facility rates were above the Australian facility rates, were:

- Item 1: An item requiring students to identify the value of coins (slightly above national)
- Item 6: An item requiring interpretation of column graphs (slightly above national)
- Item 10: An item requiring students to identify a direction on a map (slightly above national)
- Item 13: An item requiring students to locate the position of a shape after a quarter-turn (slightly above national)
- Item 14: An item requiring students to choose a suitable survey question (1.8% above national)

Items of interest, where Queensland facility rates were more than 3% below the Australian facility rates, were:

- Item 16: An item requiring students to perform a simple division (4% lower than national)
- Item 21: An item requiring students to determine one-quarter of an amount (3.7% lower than national)
- Item 22: An item requiring students to subtract a 2-digit number from a 3-digit number (4.1% lower than national)
- Item 24: An item requiring students to perform a two-step calculation involving multiplication and division (3.3% lower than national)
- Item 26: An item requiring students to add litre and millilitre amounts together (3.6% lower than national)
- Item 31: An item requiring students to select two shapes with the same area (3.9% lower than national)
- Item 34: An item requiring students to represent a word problem as a number sentence (3.2% lower than national)
- Item 38: An item requiring students to find multiple factors of a two-digit number (3.2% lower than national)

In previous NAPLAN Numeracy tests, it has been relatively common for students to skip, or not answer items, in the latter part of the test. These items are generally more challenging and require students to either provide their own constructed responses or to select **all** possible options.

In 2018, the items with the highest omission rates were Items 36, 38, 39 and 41 (all 6%) and the final item, 42, had the highest omission rate of 11%. This quite difficult Number and Algebra item required students to calculate the cost of a dozen pies after a 25% discount was applied.

The more challenging items in the Numeracy test provided students with opportunities to apply more developed skills and strategies that differentiate among the proficient students.

It is interesting to observe that in all twelve items that required constructed responses, Queensland students slightly underperformed when compared to the national cohort. Some students may lack the confidence to commit to numerical answers of their own making.

Implications for teaching

Test-wiseness

It is to be expected that teachers would instruct students to provide answers for all multiple-choice items. Even if this involves guessing some answers before the test time runs out, it is better practice to attempt these items rather than not answering at all. An educated guess may sometimes be correct.

As well as the multiple-choice questions, students should expect constructed-response items to be included in the test. They need to have the confidence to commit to providing their own numerical answers. Students can develop their mathematical confidence by regularly engaging with a wide variety of tasks that require them to produce numerical responses, either independently or in groups. A benefit could be improved response rates on standardised test items requiring constructed responses.

Problem solving

Students need frequent exposure to challenging problems to develop the confidence and skills to find the solutions. Teachers should consider teaching and learning through presenting students with problems that range from simple to complex, familiar to unfamiliar and single-step to multi-step. Problem solving should be incorporated into learning activities, not treated as a separate or add-on activity. It should be included in a range of learning experiences, both in real-life and purely mathematical contexts. Where possible, teachers are encouraged to pose maths problems to students that are derived from different curriculum areas. It is possible to equip students with approaches to apply when a numeracy problem is encountered. This will give them confidence to attempt and develop solutions.

When a problem is encountered, students should be equipped with problem-solving approaches where they feel enabled to:

- Identify the problem: What am I being asked to do?
- Analyse the problem: What do I have to work with?
- Take appropriate action: Select and apply procedures to solve the problem.
- Reflect on the answer: Check my answer — Does my solution work? Is my answer reasonable? Are there other correct answers?

Students could follow a storyline, strategise, visualise what the expression could look like and evaluate the reasonableness of the answer before looking at the available options.

While all steps are important; 'Does my solution work?' and 'Is my answer reasonable?' are particularly beneficial in test situations, especially when students are asked to construct a response.

Students should be habitually asked to consider the reasonableness of their answers. See for example Items 16, 22, 23 and 30 where the constructed responses could readily be checked for reasonableness. Knowing a problem-solving strategy will help when students are required to provide constructed responses. The items with the highest omission rates in the 2018 Numeracy test were those involving constructed responses.

Tables, graphs, diagrams

Questions that involved diagrams, pictures, tables, graphs or maps comprised 62% of the 2018 Year 5 Numeracy test.

Therefore, it is important that students are given opportunities to develop their visual literacy skills through frequent practice with problems or scenarios that involve visuals. For example, it is expected that students be familiar with geometrical representations of 2D shapes and 3D objects from different perspectives and orientations. The ability to interpret maps or simple grids to identify directions, locations and distances would also be expected. There are times when number patterns are presented in diagrammatic form, e.g. Item 17, about the number of blocks required to construct objects of increasing size.

Visual literacy in mathematics also relates to data displays or graphical displays covering varied contexts. Common examples that students should be familiar with are tally tables, column graphs, picture graphs, data tables, number lines and calendars. Students need to understand how to read simple maps to be able to identify directions, locations and distances.

There are times when pictures are used only to complement the information contained in the question stem. At other times, a picture contains the essential numerical information. Either way, it is imperative that students become familiar with pictures in the context of mathematical problems or scenarios.

Word problems

In the 2018 Year 5 Numeracy test, 16 of the 42 items (38%) were word problems that did not include visual features. Generally, word problems represent number-based scenarios in sentences. Word problems are commonly used in Statistics and Probability questions where events or samples are described verbally. Finally, word problems often describe money purchases involving change, discounts or best buys.

When dealing with a word problem, students should be encouraged to read the whole item more than once, the first time to get a general idea of what it is about and thereafter to identify important information and then to understand the actual task to be done or problem to be solved. Circling, underlining or highlighting key information may help to identify important details for students. Sorting written information into a more useful form such as drawing a sketch or diagram, a table or a list can be a valuable habit for students to adopt. The ability to translate word problems into arithmetical statements is also an important element in solving some word problems.

Difficult items

Teachers are encouraged to closely inspect the items that Queensland students found the most difficult, where the facility rates were below 30%. Teachers should consider whether it was the content covered and/or the style of these questions that made them difficult and what the implications are for their students.

QCAA resources

Full analysis of student performance and error patterns for each item is published in the SunLANDA program: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/sunlanda/accessing-navigating-sunlanda> and as PDF documents: <https://www.qcaa.qld.edu.au/p-10/naplan/test-reporting-analysis/test-item-analysis>. A school BIC and password is required to access each year level document.

A useful reference for the teaching of spatial reasoning and geometric properties is given here:

QSA, 2005, *Mathematics: About space*,
https://www.qcaa.qld.edu.au/downloads/p_10/kla_maths_info_space.pdf.